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Detection of Bilateral Tonic-Clonic Seizures Using Miniaturized Wearable Electromyography-Accelerometry Sensors

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***Abstract*— *Objective:* This study aimed to develop a seizure detection algorithm using surface electromyography (sEMG) and accelerometry (ACC) signals recorded with miniaturized wearable sensors. *Methods:* Continuous sEMG-ACC signals were acquired from patients wearing eight sensors positioned bilaterally on the upper trapezius, anterior deltoid, biceps brachii, and tibialis anterior muscles. We trained an extreme gradient boosting classifier to identify seizure epochs using setups with eight, two, and one sensor(s). Performance was evaluated via patient-wise nested cross-validation, and specificity was further assessed on an independent patient cohort without seizures. *Results:* Eleven generalized tonic-clonic seizures (GTCS) and focal to bilateral tonic-clonic seizures (FBTCS) were recorded from nine patients over 1359.6 h. The best results were obtained with a dual-sensor setup combining data from the right biceps brachii and the left tibialis anterior, achieving 100% sensitivity, 0.12 FAR/24h, and median detection latency of 22 s. On 1744.18 h of data from 19 patients without seizures, FAR/24h was 0.06. *Conclusion:* The developed algorithm effectively detected GTCS and FBTCS in an epilepsy monitoring unit, even with a reduced number of sensors. *Significance:* This approach could enable timely interventions in outpatient settings, potentially improving safety and independence for people with epilepsy.**

***Index Terms*— Accelerometry, electromyography, epilepsy, machine learning, seizure detection, wearables**

I. Introduction

EPILEPSY is a chronic neurological condition that affects ∼50 million people worldwide, with an estimated five million new cases diagnosed each year [1]. People with epilepsy (**PWE**) suffer from unpredictable and recurrent seizures due to excessive and abnormal synchronization of neuronal discharges [2]. The first line of treatment is long-term anti-seizure drug therapy; however, 30% of patients are drug-resistant and even well-controlled patients can experience breakthrough seizures if they forget one or more doses [3], [4].

Uncontrolled seizures are associated with cognitive impairment, psychosocial handicap, psychiatric comorbidities, and up to three times the standard mortality rate due to seizure-related injuries [5]. Patients with poorly controlled seizures are at a higher risk of having accidental injuries compared to the general population [6]. Among seizure types, generalized tonic-clonic seizures (**GTCS**) and focal to bilateral tonic-clonic seizures (**FBTCS**) are particularly associated with a higher risk of burns, fractures, dislocations, and head or dental trauma [7]. Moreover, they are the primary risk factor for sudden unexpected death in epilepsy (**SUDEP**) [8], [9].

Rapid intervention during or immediately following a seizure can reduce the risk of injuries and fatal outcomes. Evidence suggests that early peri-ictal intervention [10]—or, in cases of seizure-induced cardiorespiratory arrest, an intervention within three minutes [11]—could reduce the risk of SUDEP, highlighting the importance of timely seizure detection. The gold standard to detect epileptic seizures is video-electroencephalography (**EEG**). However, this is typically confined to inpatient settings, such as an epilepsy monitoring unit (**EMU**). This limitation has driven interest in wearable seizure detection devices, suitable for everyday use and able to alert caregivers for timely assistance during seizures.

Over the last years, there has been extensive research on seizure detection wearables, especially those targeting GTCS. Such devices rely on physiological signals from which biomarkers of seizure-like activity can be identified. For instance, Beniczky *et al*. [12] evaluated Seizure Link (formerly known as EDDI), a dedicated seizure detection device based on surface electromyography (**sEMG**) recorded from the biceps brachii. Similarly, Halford *et al.* [13] assessed the SPEAC system, another sEMG-based biceps monitor. Other seizure detection wearables include wrist-worn devices such as EpiCare [14], which uses accelerometry (**ACC**) data, and Embrace [15], which combines ACC and electrodermal activity (**EDA**) to detect GTCS. More recently, Gharbi *et al*. [16] explored the potential of a biometric shirt to detect FBTCS based on electrocardiography (**ECG**) and ACC data.

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Despite these advances, the application of miniaturized technology in seizure detection remains largely unexplored. Smaller, easily concealable devices could help reduce the stigma often associated with wearing medical equipment. The optimal placement of sEMG-based seizure detection devices also remains understudied. Most studies have focused on the biceps brachii [12], [13], [17], with one exploring the deltoids and tibialis muscles [18]. Additionally, a persistent challenge in many wearable systems is the high false alarm rate, which can lead to alarm fatigue and eventual device abandonment.

In this study, we developed a seizure detection algorithm using miniaturized wearable sEMG-ACC sensors. The algorithm was trained on continuous recordings from patients in an EMU, each wearing eight sensors placed on muscles commonly involved in the motor manifestations of convulsive seizures. Performance was evaluated with two methods: **1)** a nested cross-validation across patients with ≥ 1 recorded GTCS/FBTCS, and **2)** evaluating false alarm rate on a different patient cohort with no recorded clinical epileptic seizures. Furthermore, we used these two strategies to evaluate the algorithm's performance when reducing the number of sensors.

II. Materials and Methods

A. Study Design

All patients were recruited prospectively at the EMU of the University of Montreal Hospital Center (**CHUM**) between September 2023 and August 2025. Inclusion criteria were a confirmed diagnosis of epilepsy and a documented history of motor seizures. Patients with implanted devices, such as a pacemaker, were excluded from recruitment. This study was approved by the CHUM's Research Ethics Board (18.091), and written informed consent was obtained from all patients or their legal representatives prior to enrollment.

B. Data Acquisition

1) Acquisition System: Continuous sEMG-ACC signals were acquired using Cometa's Wave Plus and picoEMG sensors [19]. Fig. 1 shows the acquisition setup at the EMU. Each sensor (Fig. 1(a)) contains a triaxial accelerometer, weighs seven grams, measures 41 × 16 × 8.5 mm, and has two built-in clip connectors for electrode attachment without requiring a separate ground electrode. Standard pre-gelled Ag/AgCl electrodes were used for all recordings in this study. The sEMG and ACC signals were collected at a sampling frequency of 2kHz and 142.8 Hz, respectively. Due to their 10-hour battery life, the sensors were replaced with fully charged sets three times daily, at ∼07:30, 14:30, and 21:30. The charging time to achieve full battery was eight hours. Sensor replacement was performed by the research team (ISW, DAGL, AJ, SC, RH, CRu, OG, MR, AP, CRo) to ensure 24-hour recordings and to verify correct sensor placement throughout the study.

2) Sensor Placement: All recruited patients wore eight sensors during their stay at the EMU, while undergoing simultaneous video-EEG monitoring (Fig. 1(d)-(e)). The sensors were placed symmetrically on the belly of the following muscles: left and right upper trapezius, anterior deltoid, biceps brachii, and tibialis anterior (Fig. 2(a)). An example of the sensors placed on both biceps brachii is shown in Fig. 2(b). The anterior deltoid, biceps brachii, and tibialis anterior were selected due to their involvement in GTCS/FBTCS [12], [13], [17], [18], [20], and to capture both upper and lower limb activity during seizures. The upper trapezii were included to record activity proximal to the neck, as head deviations are a common early semiological sign in some types of epilepsy (e.g., temporal and frontal lobe epilepsy) [21]. Placement on the sternocleidomastoid was avoided due to its susceptibility to head-motion artifacts and possible patient discomfort.

C. Seizure Annotations

Seizures were annotated by three board-certified epilepsy fellows (JPMS, GD, AR) based on video-EEG recordings and completely blinded to the sEMG-ACC signals. They annotated the seizure onset and offset times, including electrographic, clinical, motor, and generalization (i.e., the time of progression to the bilateral tonic-clonic phase) onsets, and the type of seizure according to the ILAE 2025 Classification of Seizure Types [22]. Fig. 2(c) and 2(d) show examples of annotated sEMG and ACC signals, respectively, during an FBTCS.

D. Signal Preprocessing

To reduce motion artifacts, sEMG signals were filtered with a 2nd-order Butterworth high-pass filter (2 Hz) and a 60 Hz notch filter to remove powerline interference, then downsampled to 1kHz. ACC signals were filtered with a 2nd-order Butterworth low-pass filter (20 Hz) to reduce artifacts from abrupt impacts or instability due to loose electrodes.

E. Feature Extraction

Features were extracted from the sEMG-ACC signals using an overlapping sliding window. This approach was used to prepare the data for machine learning analysis by dividing the continuous time-series into shorter, overlapping segments (i.e. windows) of fixed duration. Previous research on sEMG-based seizure detection has used 1- to 3-second windows [12], [18], [23], whereas studies incorporating other modalities (e.g., ECG, ACC) often use larger windows ranging from 10 to 30 seconds [16], [24]. Based on these values, we generated four separate feature datasets using the following configurations: 3-second sliding window with 50% overlap, 5-second sliding window with 50% overlap, 8-second sliding window with 75% overlap, and 10-second sliding window with 75% overlap.

1) sEMG Features: Fifteen features were extracted from each sEMG segment across all muscles, totaling 236 sEMG-derived features. Five were extracted from the time domain, including root mean square (**RMS**), zero-crossing rate (**ZCR**) with a $\pm 50\mu V$ threshold [18], variance, mean absolute value (**MAV**), and the integrated EMG, yielding 40 temporal features (5 features × 8 muscles). Four frequency-domain features were derived using Welch's method for power spectral density estimation [25]: median frequency, relative power in the 100–500 Hz band [23], magnitude-squared coherence between bilateral muscle pairs, and peak frequency. This contributed 28 spectral features (3 features × 8 muscles + 1 feature × 4

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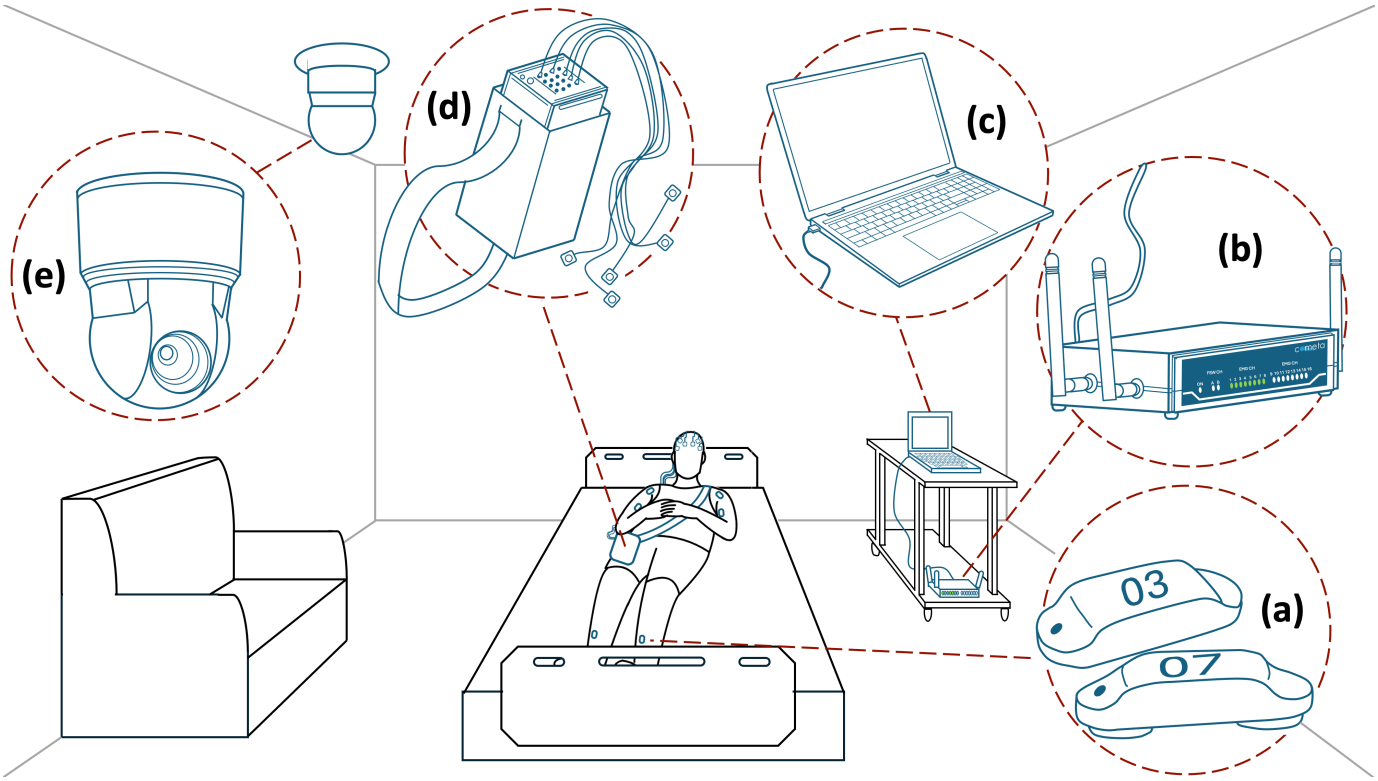

Fig. 1. Data acquisition setup in the epilepsy monitoring unit. Patients wore eight multimodal picoEMG sensors **(a)** that wirelessly transmitted data to the Wave Plus receiving unit **(b)**. The unit was connected via USB cable to a computer **(c)** running Cometa's EMG and Motion Tools software, which was used to assign each sensor to a specific muscle, select the appropriate radio frequency channel for data transmission, and prospectively store the data in C3D format. At the same time, patients were continuously monitored with simultaneous electroencephalography **(d)** and ceiling-mounted video recording **(e)**.

muscle pairs). Furthermore, the Discrete Wavelet Transform with a Daubechies–4 wavelet was applied to decompose each signal into four frequency bands: 2–8 Hz, 8–62 Hz, 62–250 Hz, and 250–500 Hz. Five features were extracted from each band's resulting coefficients: band power, entropy, standard deviation (**STD**), skewness, and kurtosis. Additionally, the high-to-low frequency ratio [20] was computed as the ratio of band power in the 62–250 Hz band to that in the 2–8 Hz band. This resulted in 168 wavelet features (5 features $\times$ 4 bands $\times$ 8 muscles + 1 feature $\times$ 8 muscles).

2) ACC Features: Twelve features were extracted from each 3D ACC segment across all muscles, totaling 384 ACC-derived features. For each ACC axis (*x, y, z*) and signal norm, nine time-domain features were computed: RMS, ZCR, STD, variance, MAV, mean jerk as the rate of change of acceleration, maximum jerk, STD of jerk, and interquartile range. This resulted in 288 temporal features (9 features $\times$ 4 axes and norm $\times$ 8 muscles). Three frequency-domain features were extracted from each ACC axis and signal norm using Welch's method [25]: median frequency, relative power in the 0–5 Hz band for human motion, and peak frequency. This added 96 spectral features (3 features $\times$ 4 axes and norm $\times$ 8 muscles).

3) Early Fusion and Labeling: Prior to classification, the sEMG- and ACC-derived features were concatenated to form a single feature vector per time segment. This resulted in a feature matrix with 620 columns (236 sEMG + 384 ACC features), where each row represented one time-segment. Each segment was assigned a binary label indicating whether it corresponded to a seizure based on the annotated motor onset and offset times (see Section II-C). A segment was labeled as 1 (*seizure*) if it overlapped with a true seizure annotation; otherwise, it was labeled as 0 (*non-seizure*).

F. Development of Machine Learning Algorithm

1) Classification: An extreme gradient boosting (**XGBoost**) [26] classifier was trained to identify seizure segments from non-seizure segments. We explored different values for the *maximum tree depth* and *minimum child weight* (see Table S1), hyperparameters that control the complexity of the trees and the number of instances required in a branch to allow further splitting, respectively. Increasing the maximum depth allows the model to learn more detailed patterns but can lead to overfitting. In contrast, increasing the minimum child weight helps prevent overfitting by avoiding splits based on very few data points. Given the high dimensionality of the feature space, we adopted conservative strategies to avoid overfitting. We increased the *gamma* (i.e., minimum loss reduction) hyperparameter to promote simpler trees unless a split yielded a meaningful gain. Additionally, we applied both L_1 and L_2 regularization, which encourage sparsity and discourage overly complex models, respectively. In particular, the L_1 regularization acts as an embedded feature selector; though it does not explicitly reduce the input dimension, it drives the coefficients of irrelevant and low-contributing

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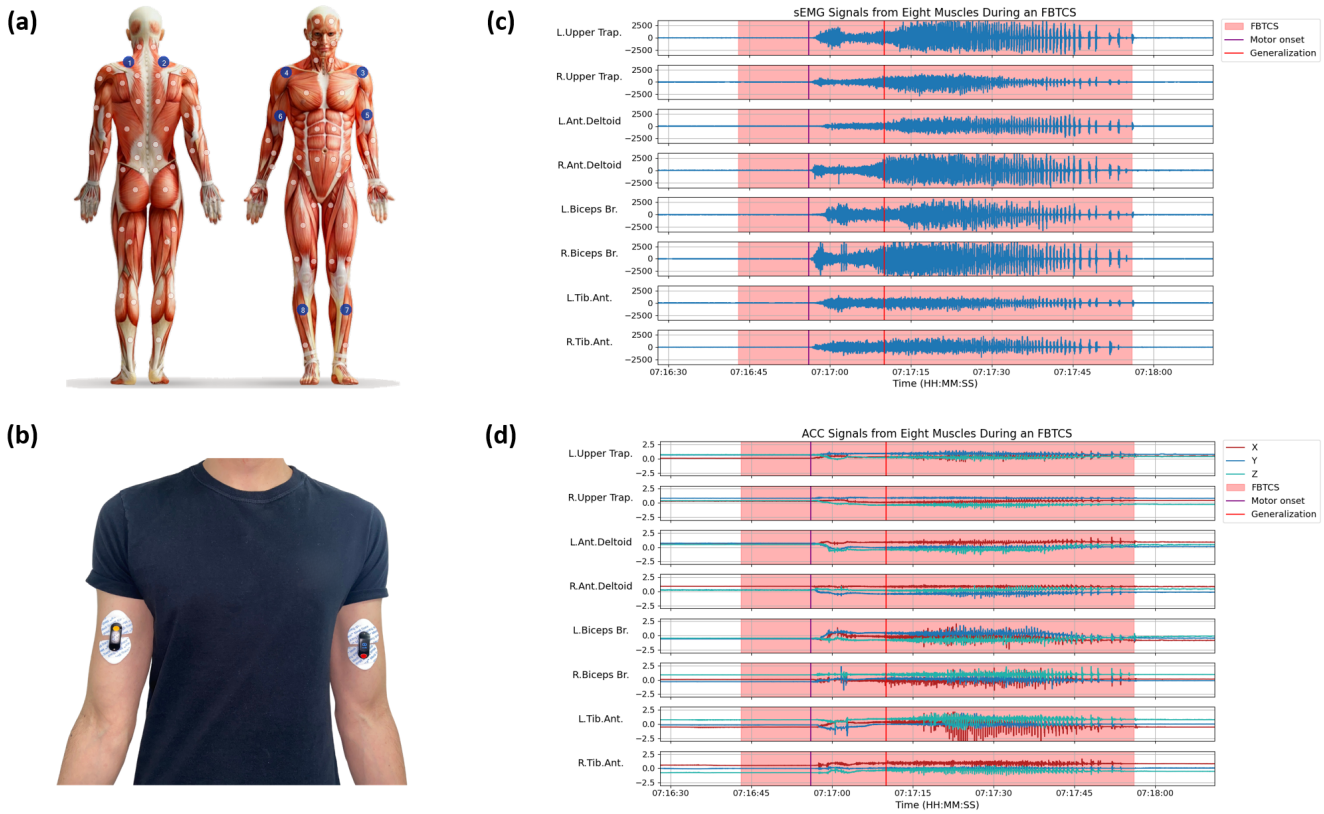

Fig. 2. **(a)** Sensor placement on the body: (1) left upper trapezius, (2) right upper trapezius, (3) left anterior deltoid, (4) right anterior deltoid, (5) left biceps brachii, (6) right biceps brachii, (7) left tibialis anterior, (8) right tibialis anterior. **(b)** An example of two picoEMG sensors placed on the biceps brachii muscles. **(c)** Surface electromyography signals recorded from eight muscles during a focal to bilateral tonic-clonic seizure (FBTCS). The red region marks the electrographic seizure onset and offset, the vertical purple line marks the motor onset, and the red vertical line marks the generalization as annotated by an epilepsy fellow based solely on the video-EEG reference. **(d)** Triaxial accelerometry signals recorded from eight muscles during the same FBTCS illustrated in (c).

features to zero during training [27]. Due to the scarce nature and short duration of seizures, the dataset was highly imbalanced. Therefore, we used the XGBoost *weight scaling* hyperparameter to increase the influence of the positive class (i.e., seizure segments) during training. We also increased the *maximum delta step* to help stabilize learning by preventing trees from making overly confident predictions in a single step.

2) Postprocessing: To reduce the number of isolated false positives, the classifier's output was postprocessed using the firing power method [28]. This approach consists of two main steps: **1)** regularization, where a moving average with a predefined window size τ is computed over the classifier's binary output; and **2)** thresholding, which triggers an alarm whenever the average exceeds a predefined threshold L.

G. Evaluation of the Machine Learning Algorithm

1) Performance Metrics: To provide clinically relevant metrics, the performance of the algorithm was ultimately assessed at the *event* level. Events were defined by grouping positive-classified segments that occurred within 60 seconds of each other. This merging window was chosen to account for potential residual motor activity—such as sustained clonic jerks or muscle spasms—that may persist beyond the electrographic seizure offset. Events were considered true positives (**TP**) if they overlapped, however minimally, with a reference label. No pre-ictal or post-ictal tolerance was applied in this study. From the count of TP, false negative (**FN**), and false positive (**FP**) events, we derived: **1)** sensitivity, **2)** false alarm rate (**FAR**) per 24 hours, **3)** time in false warning (**TiW**), calculated as the duration of each false alarm in seconds; and **4)** detection latency, measured in seconds relative to the electrographic, motor, and generalization onset times. Since FAR can vary substantially between patients and recording durations, both micro and macro FAR are reported as follows:

$$MicroFAR = \frac{\sum_{i=1}^{N} FP_i}{\sum_{i=1}^{N} T_i} \times 24, \tag{1}$$

where N is the total number of patients, FP_i is the number of false alarms for patient i, and T_i is the total device recording time for patient i in hours. This measure pools all false alarms and recording times across patients before calculation, reflecting the overall dataset-level FAR.

$$MacroFAR = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{FP_i}{T_i} \times 24 \right), \tag{2}$$

where the FAR is computed individually for each patient and then averaged across patients, giving them equal weight regardless of recording duration.

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2) Nested Cross-Validation: To optimize model performance without introducing data leakage, we implemented a patient-wise nested cross-validation (**NCV**) for each feature dataset. We used a 9-fold outer loop (leave-one-patient-out, as advised for small datasets [29]) for model evaluation and a 3-fold inner loop for model selection. In each outer fold, one patient was held out for testing, while the rest were split into training (70%) and validation (30%) sets (see Fig. S1). Hyperparameters for both the classifier and postprocessing were optimized based on the highest average sample-based F1-score across the validation sets in the inner loop, ensuring that optimal models were always selected based on their performance across different partitions of unseen data. Furthermore, to address class imbalance without compromising the integrity of out-of-sample testing, the XGBoost *weight scaling* hyperparameter was set dynamically based on the class distribution within each training split. Specifically, it was computed as the ratio of negative to positive instances in each training set.

3) Evaluation on Non-Seizure Cohort: The algorithm was further tested on patients who did not have any recorded clinical epileptic seizures. We refer to this group as the *non-seizure cohort*, whereas patients included in the NCV are referred to as the *seizure cohort*. Using the same model selection strategy as in the NCV's inner loop, the optimal hyperparameters were identified via 3-fold cross-validation on the seizure cohort. A final model was then trained on the full seizure cohort and evaluated on the non-seizure cohort. The goal was to provide an additional estimate of the model's FAR in a distinct sample with a larger number of patients.

4) Performance Evaluation Using One Sensor: To explore the impact of using fewer sensors on our algorithm's performance, we trained the XGBoost model with data from one muscle at a time, using the same hyperparameter grid-search (Table S1) and evaluation strategy as with the 8-sensor setup.

III. Results

A. Patient Cohort

Fifty patients were initially recruited for this study. However, due to a synchronization problem at the EMU between March and May 2024, two patients were excluded. During this period, the clock of the video-EEG system exhibited a non-linear drift from the Coordinated Universal Time, resulting in a misalignment of the sensor data with the EEG. Consequently, the data from these two patients could not be reliably referenced to the gold standard. Importantly, these two patients did not experience FBTCS/GTCS during their participation. The study's inclusion flow is summarized in Fig. S2.

Among the remaining 48 patients, ten had at least one GTCS or FBTCS while wearing the sensors; however, two seizures were not recorded due to battery limitations, leaving 11 seizures from nine patients available for analysis. These nine patients were included in the seizure cohort, as detailed in Table I. One of the unrecorded seizures occurred in a patient (p4) who had two other FBTCS, both of which were included in the analysis. The other unrecorded GTCS occurred in a patient (p17) who had no other clinical epileptic seizures; therefore, this patient was included in the non-seizure cohort. Three patients from the seizure cohort also had other types of seizures, including focal impaired consciousness seizures (**FICS**) and generalized seizures (**GS**). The majority of patients in this cohort were female (67%), and the median age was 33 years (range: 19-45 years). The total device recording time was 1,359.6 h, with an average of 151.06 h per patient (median: 135.49 h, range: 67.87–340.11 h). GTCS and FBTCS lasted an average of 106.9 s (median: 118 s, range: 62–140 s), totaling 19.6 min and accounting for 0.02% of the total recording time. More than one FBTCS was recorded in only two patients.

Sensor detachment, defined as either **1)** one or more sensors unclipping from one or both electrodes or **2)** one or more electrodes losing adhesion to the skin, occurred during four of the 11 recorded seizures. As a result, seizure-like activity in these cases was captured by only four to seven sensors instead of the full set of eight. In one case, two sensors lost power prior to seizure onset, and the subsequent GTCS was recorded by only six active sensors. However, to reflect real-world conditions where such scenarios may occur, none of these seizures were excluded from the analysis.

During the acquisition period, 19 patients had no clinical epileptic seizures recorded by the sensors, and were therefore included in the non-seizure cohort (see Table I). Within this group, two patients had psychogenic non-epileptic seizures (**PNES**), totaling two recorded PNES. This cohort also consisted predominantly of females (68%), and the median age was 33 years (range: 20-65 years). The total device recording time was 1,744.18 h, with an average of 91.8 h per patient (median: 66.98 h, range: 19.04–502.75 h).

B. Performance Using Full 8-Sensor Setup

1) Seizure Cohort: The results obtained using different feature extraction parameters (sliding window length, overlap percentage) are summarized in Table II. The highest performance (**90.9% sensitivity**) was achieved when the algorithm was trained on features extracted using a 5-second sliding window, detecting **10/11 seizures**, with both micro and macro FAR/24h of **0.05** (median across patients: **0**, range: 0 – 0.27). For this model, the median TiW was **5 s** (range: 5–87.5 s), and median latencies were **60.25 s** (range: 26–97.5 s), **38 s** (range: 10.5–73.5 s), and **22.5 s** (range: -1–49.5 s) from the electrographic, motor, and generalization onset times, respectively. False alarms only occurred in three patients (33.3%). The longest (87.5 s) was produced by high muscle activity in the form of rhythmic voluntary movements. The patient was off-camera during this period, but subsequent review confirmed that the event was unrelated to seizure activity. No alarms were triggered by the other recorded focal and GS.

Interestingly, the only FBTCS that was missed by the best-performing model (see Fig. 3(a)) coincided with both trapezius sensors detaching during the seizure, suggesting that these muscles might have an important contribution to the algorithm's prediction. This seizure was consistently not detected by any of the other models (Fig. S3(a), (c)-(d)). However, when using 8- and 10-second sliding windows, the algorithm detected the seizure six and 8.5 seconds after its electrographic offset, respectively. The seizure was therefore missed, and this event was counted as a false alarm.

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TABLE I
PATIENT DEMOGRAPHICS

	Patient	Sex	Age, years	Epileptic foci[a]	Total recording time, h	Recorded GTCS/FBTCS, n	Other recorded seizures, n
Seizure cohort	1	F	36	Bitemporal	85.78	1 (FICS-FBTCS)	-
	2	F	45	Generalized	67.87	1 (GTCS)	90 (GS)
	3	F	24	Generalized	85.06	1 (GTCS)	-
	4	M	19	Left mid-posterior temporal	340.11	2 (FBTCS)	-
	5	M	21	Bitemporal	141.44	1 (FBTCS)	-
	6†	F	36	Left temporal	254.30	1 (FBTCS)	-
	7	F	33	Right temporal plus	167.48	2 (FICS-FBTCS)	1 (FICS)
	8†	F	43	Bitemporal	135.49	1 (FBTCS)	2 (FICS)
	9	M	33	Left temporal	82.07	1 (FBTCS)	-
	Total	**6/3 (F/M)**	**-**	**-**	**1359.6**	**11**	**-**
	Median [range]	**-**	**33 [19 - 45]**	**-**	**135.49 [67.87 - 340.11]**	**1 [1 - 2]**	**-**
Non-seizure cohort	10	F	23	Bilateral	39.44	-	-
	11	M	37	Right central	28.04	-	-
	12	M	25	Generalized	48.74	-	-
	13	F	21	Left frontal central	94.58	-	1 (PNES)
	14	F	44	Right temporal	105.68	-	-
	15	F	30	Right central	66.98	-	-
	16	F	33	Left temporal	35.76	-	-
	17	F	21	Juvenile Myoclonic Epilepsy	89.39	-	-
	18	F	65	Generalized	130.69	-	-
	19	M	33	Bitemporal	25.53	-	-
	20†	M	40	Bilateral multifocal	87.28	-	1 (PNES)
	21	F	20	Left temporal	167.24	-	-
	22	F	20	Generalized	86.47	-	-
	23	F	38	Left frontal	19.04	-	-
	24	M	57	Generalized	60.72	-	-
	25	M	23	Generalized	70.88	-	-
	26	F	47	Left frontal	21.94	-	-
	27	F	21	Left temporal	63.02	-	-
	28	F	36	Left temporal	502.75	-	-
	Total	**13/6 (F/M)**	**-**	**-**	**1744.18**	**-**	**-**
	Median [range]	**-**	**33 [20 - 65]**	**-**	**66.98 [19.04 - 502.75]**	**-**	**-**

Abbreviations: Female (F), male (M), generalized tonic-clonic seizure (GTCS), focal to bilateral tonic-clonic seizure (FBTCS), focal impaired consciousness seizure (FICS), generalized seizure (GS), psychogenic non-epileptic seizure (PNES).
[a]Epileptic foci were not validated by intracranial EEG in all patients.
†Patients who underwent intracranial EEG during their participation in the study.

2) Non-Seizure Cohort: Table III summarizes the results of all models across the non-seizure cohort, and the final selected hyperparameters are detailed in Table S2. The lowest micro FAR/24h (**0.08**) was achieved by the model trained on features extracted using a 3-second sliding window, where false alarms only occurred in four patients (21%). With the best-performing configuration from the seizure cohort (5-second sliding window), the model achieved both micro and macro FAR/24h of **0.56** (median across patients: **0**, range: 0 – 4.56). This model generated false alarms in eight patients (42.1%), and two of these patients (p13, p20) accounted for 68% of all false alarms (see Fig. 3(a)). Only the model trained on features extracted using an 8-second sliding window detected one of the two PNES, which was counted as a false alarm.

C. Performance Using Reduced Sensor Setups

1) Seizure Cohort: To assess whether performance could be preserved with minimal setups, the algorithm was trained on data recorded exclusively from the right biceps brachii (**RB**) and the left tibialis anterior (**LTi**), both separately and jointly (**RB + LTi**). These muscles were selected to enable a fair comparison with the 8-sensor setup, as these were the only sites where the sensors remained attached during all seizures, thereby including the same number of seizures (n = 11) in the analysis. Table IV summarizes the results for each feature extraction configuration and sensor setup. Additionally, the same analysis was conducted for the other muscles, separately, to provide an estimate of the algorithm's performance by using the available number of seizures in each case (i.e., the total number of seizures recorded by each individual sensor). The corresponding results are reported in Tables S5–S8.

With all three sensor setups (RB, LTi, and RB + LTi), the algorithm achieved a sensitivity of **100%**, correctly detecting all 11 seizures regardless of the feature extraction configuration. The highest performance was achieved with the 2-sensor setup, combining information from both the RB and LTi muscles, specifically when features were extracted using a 3-second sliding window. This model achieved a micro FAR/24h of

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TABLE II
8-SENSOR SETUP PERFORMANCE ON THE SEIZURE COHORT UNDER DIFFERENT FEATURE EXTRACTION CONFIGURATIONS

Performance Metric	SW = 3 s, OVLP = 50%	SW = 5 s, OVLP = 50%	SW = 8 s, OVLP = 75%	SW = 10 s, OVLP = 75%
Sensitivity, % [detected/total]	81.8 [9/11]	**90.9** [**10**/11]	81.8 [9/11]	81.8 [9/11]
Micro FAR/24h [total false alarms]	0.22 [13]	**0.05** [**3**]	0.30 [17]	0.14 [8]
Macro FAR/24h [median, range]	0.25 [0.28, 0 – 0.58]	**0.05** [**0**, 0 – 0.27]	0.31 [**0**, 0 – 1.52]	0.12 [**0**, 0 – 0.50]
Median TiW, s [range]	3 [0 – 69]	5 [5 – 87.5]	4 [4 – 76]	5 [5 – 97.5]
Median electrographic latency, s [range]	**59** [21 – 88.5]	60.25 [26 – 97.5]	61 [22 – 96]	65 [26 – 102.5]
Median motor latency, s [range]	**34** [8 – 72]	38 [10.5 – 73.5]	36 [9 – 72]	41 [10.5 – 76]
Median generalization latency, s [range]	**16** [-6 – 30]	22.5 [-1 – 49.5]	20 [-5 – 28]	22 [-1 – 54.5]

The best results for each row are presented in bold. Abbreviations: sliding window (SW), overlap (OVLP), false alarm rate (FAR), time in false warning (TiW).

TABLE III
8-SENSOR SETUP PERFORMANCE ON THE NON-SEIZURE COHORT UNDER DIFFERENT FEATURE EXTRACTION CONFIGURATIONS

Performance Metric	SW = 3 s, OVLP = 50%	SW = 5 s, OVLP = 50%	SW = 8 s, OVLP = 75%	SW = 10 s, OVLP = 75%
Micro FAR/24h [total false alarms]	**0.08** [**6**]	0.56 [41]	1.1 [80]	0.55 [40]
Macro FAR/24h [median, range]	**0.06** [**0**, 0 – 0.45]	0.56 [**0**, 0 – 4.56]	1.12 [**0**, 0 – 8.12]	0.61 [**0**, 0 – 3.04]
Median TiW, s [range]	6 [3 – 10.5]	7.5 [0 – 60]	6 [0 – 90]	5 [0 – 72.5]

The best results for each row are presented in bold. The results obtained with the best-performing configuration from the seizure cohort are presented in blue. Abbreviations: sliding window (SW), overlap (OVLP), false alarm rate (FAR), time in false warning (TiW).

0.12 and macro FAR/24h of **0.14** (median across patients: **0**, range: 0 – 0.71). The median TiW was **1.5 s** (range: 0–10.5 s), and median latencies were **59 s** (range: 19–96 s), **34 s** (range: 8.5–84 s), and **22 s** (range: -4.5–44 s) from the electrographic, motor, and generalization onset times, respectively. Figure 3(d) shows the model's sensitivity and FAR/24h per patient.

Between the RB and LTi muscles, the latter achieved higher performance, with the best model (10-second sliding window) achieving micro and macro FAR/24h of **0.28** and **0.34** (median across patients: **0**, range: 0–1.75), respectively. The performance results per patient are shown in Figure 3(c). For this model, median TiW was **7.5 s** (range: 0–145 s), and median latencies were **65 s** (range: 22–107 s), **40 s** (range: 14–73.5 s), and **26.5 s** (range: 6.5–57 s) from the electrographic, motor, and generalization onset times, respectively. In contrast, the best RB-based model (8-second sliding window) achieved a micro FAR/24h of **0.51** and macro FAR/24h of **0.48** (median across patients: **0.28**, range: 0–2.57). Median TiW was **8 s** (range: 2–146 s), and median latencies were **74 s** (range: 27–100 s), **44 s** (range: 13–88 s), and **32 s** (range: 1–51 s) from the electrographic, motor, and generalization onset times, respectively. Figure 3(b) shows this model's sensitivity and FAR/24h per patient. Although there was an increase in FAR compared to the other setups, a single patient (p7) accounted for the majority of false alarms (62%).

Notably, the same event that produced the longest (87.5 s) false alarm with the 8-sensor setup also triggered long alarms (66–145 s) with both single-sensor setups, and was the cause of the longest alarm (10.5 s) generated by the best model in the 2-sensor setup. None of the best-performing models from these three setups raised any alarms during the other recorded focal and GS. The results per patient when using different feature extraction parameters are shown in Fig. S4–S6.

2) Non-Seizure Cohort: The results of the three sensor setups across the non-seizure cohort are summarized in Table V, and the final selected hyperparameters are detailed in Tables S3–S4. As with the seizure cohort, the lowest micro FAR/24h (**0.06**) was achieved with the 2-sensor setup, specifically by the model trained on features extracted using a 3-second sliding window. This model did not generate any false alarms in the majority of patients (78.9%).

When using RB data only, the lowest micro FAR/24h (**0.42**) was achieved by the model trained on features extracted using a 10-second sliding window. In contrast, when using LTi data only, the lowest micro FAR/24h (**0.30**) was achieved by the model trained on features extracted using a 5-second sliding window. When using the best-performing configurations from the seizure cohort (RB: 8-second sliding window; LTi: 10-second sliding window), the RB-based model achieved a micro FAR/24h of **0.50**, whereas the LTi-based model achieved **0.62**. With the latter, the same patient (p20) drove 44% of all false alarms (see Fig. 3(c)). Only the RB-based model raised a false alarm during one of the two PNES.

IV. DISCUSSION

In this study, we developed a seizure detection algorithm for GTCS and FBTCS using sEMG and ACC data recorded from PWE wearing eight miniaturized sensors during their stay in an EMU. We further assessed the algorithm's performance when restricted to single- and dual-sensor inputs. In both cases, performance was evaluated on data from held-out patients via NCV and an additional, independent patient cohort to assess specificity. To the best of our knowledge, this is the first study to investigate multimodal seizure detection using miniaturized, wireless sEMG–ACC sensors.

Given the limited number of recorded GTCS and FBTCS, this work would be classified as a Phase 1 study based on es-

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TABLE IV
PERFORMANCE OF REDUCED SENSOR SETUPS ON THE SEIZURE COHORT UNDER DIFFERENT FEATURE EXTRACTION CONFIGURATIONS

1-Sensor Setup Performance	**SW = 3 s, OVLP = 50%**		**SW = 5 s, OVLP = 50%**		**SW = 8 s, OVLP = 75%**		**SW = 10 s, OVLP = 75%**	
	RB	**LTi**	**RB**	**LTi**	**RB**	**LTi**	**RB**	**LTi**
Sensitivity, % [detected/total]	**100** [**11**/11]	**100** [**11**/11]	**100** [**11**/11]	**100** [**11**/11]	**100** [**11**/11]	**100** [**11**/11]	**100** [**11**/11]	**100** [**11**/11]
Micro FAR/24h [total false alarms]	0.70 [40]	0.33 [19]	0.58 [33]	0.41 [23]	**0.51** [**29**]	0.52 [30]	0.88 [50]	**0.28** [**16**]
Macro FAR/24h [median, range]	0.69 [0.28, 0–3.72]	0.44 [0.21, 0–2.52]	0.56 [**0.09**, 0–4.15]	0.50 [0.50, 0–1.68]	**0.48** [0.28, 0–2.57]	0.74 [0.34, 0–2.63]	0.79 [**0.09**, 0–6.44]	**0.34** [**0**, 0–1.75]
Median TiW, s [range]	6 [0–78]	16.5 [0–289.5]	7.5 [0–130]	7.5 [0–100]	8 [2–146]	5 [0–118]	6.25 [0–225]	7.5 [0–145]
Median electrographic latency, s [range]	**67** [25.5–100.5]	**61** [16–103]	72.5 [28.5–105]	72 [22–103]	74 [27–100]	70 [26–102]	70.5 [29.5–103]	65 [22–107]
Median motor latency, s [range]	**37.5** [12.5–88.5]	50 [8–76.5]	45.5 [15.5–88.5]	46 [14–87.5]	44 [13–88]	46 [13–88]	46 [18–91]	**40** [14–73.5]
Median generalization latency, s [range]	31.5 [-1.5–44]	26.5 [-3–67]	27.5 [1.5–57]	26.5 [4–67]	32 [1–51]	**26** [-1–61]	**26** [4–54.5]	26.5 [6.5–57]
2-Sensor Setup Performance	**RB + LTi**		**RB + LTi**		**RB + LTi**		**RB + LTi**	
Sensitivity, % [detected/total]	**100** [**11**/11]		**100** [**11**/11]		**100** [**11**/11]		**100** [**11**/11]	
Micro FAR/24h [total false alarms]	**0.12** [**7**]		0.14 [8]		0.35 [20]		0.35 [20]	
Macro FAR/24h [median, range]	**0.14** [**0**, 0-0.71]		0.16 [0.14, 0-0.56]		0.37 [0.29, 0-1]		0.45 [0.29, 0-1.7]	
Median TiW, s [range]	1.5 [0-10.5]		5 [0-15]		5 [2-62]		6.25 [2.5-100]	
Median electrographic latency, s [range]	**59** [19-96]		65.5 [19.5-98]		66 [21-97]		65.5 [22-102.5]	
Median motor latency, s [range]	**34** [8.5-84]		42.5 [10.5-86]		44 [11-82]		40 [13-76]	
Median generalization latency, s [range]	22 [-4.5-44]		24.5 [-3.5-44.5]		**21** [-1-54]		26 [-1-54.5]	

The best results for each row and muscle are presented in bold. Abbreviations: sliding window (SW), overlap (OVLP), false alarm rate (FAR), time in false warning (TiW), right biceps brachii (RB), left tibialis anterior (LTi).

TABLE V
PERFORMANCE OF REDUCED SENSOR SETUPS ON THE NON-SEIZURE COHORT UNDER DIFFERENT FEATURE EXTRACTION CONFIGURATIONS

1-Sensor Setup Performance	**SW = 3 s, OVLP = 50%**		**SW = 5 s, OVLP = 50%**		**SW = 8 s, OVLP = 75%**		**SW = 10 s, OVLP = 75%**	
	RB	**LTi**	**RB**	**LTi**	**RB**	**LTi**	**RB**	**LTi**
Micro FAR/24h [total false alarms]	2.27 [165]	1.1 [80]	0.71 [52]	**0.30** [**22**]	0.50 [37]	0.84 [61]	**0.42** [**31**]	0.62 [45]
Macro FAR/24h [median, range]	2.13 [1.67, 0-8.37]	1.53 [0.28, 0-14.55]	0.76 [0.33, 0-3.84]	0.64 [**0**, 0-5.99]	0.60 [**0**, 0-3.04]	0.63 [**0**, 0-7.15]	**0.48** [0.14, 0-2.46]	**0.52** [**0**, 0–5.5]
Median TiW, s [range]	4.5 [0-183]	3 [0–975]	5 [0–80]	7.5 [0–227.5]	12 [4-54]	6 [0-688]	7.5 [2.5-72.5]	7.5 [0-180]
2-Sensor Setup Performance	**RB + LTi**		**RB + LTi**		**RB + LTi**		**RB + LTi**	
Micro FAR/24h [total false alarms]	**0.06** [**4**]		0.85 [62]		0.4 [29]		0.19 [14]	
Macro FAR/24h [median, range]	**0.04** [**0**, 0-0.27]		0.8 [0.14, 0-5.77]		0.46 [**0**, 0-3.02]		0.24 [**0**, 0-1.9]	
Median TiW, s [range]	6.75 [4.5-16.5]		10 [0-92.5]		4 [0-52]		6.25 [0-10]	

The best results for each row and muscle are presented in bold. The results obtained with the best-performing configuration from the seizure cohort are presented in blue. Abbreviations: sliding window (SW), overlap (OVLP), false alarm rate (FAR), time in false warning (TiW), right biceps brachii (RB), left tibialis anterior (LTi).

tablished standards for testing and clinical validation of seizure detection devices [30]. However, aside from the sample size criterion, all other requirements for a Phase 2 study were met. Seizures were identified based on video-EEG recordings as the reference standard, with the annotation process completely blinded to the sensor data. To mitigate selection bias due to our small sample size, the algorithm was trained on continuous data from PWE, using all the recordings from every patient

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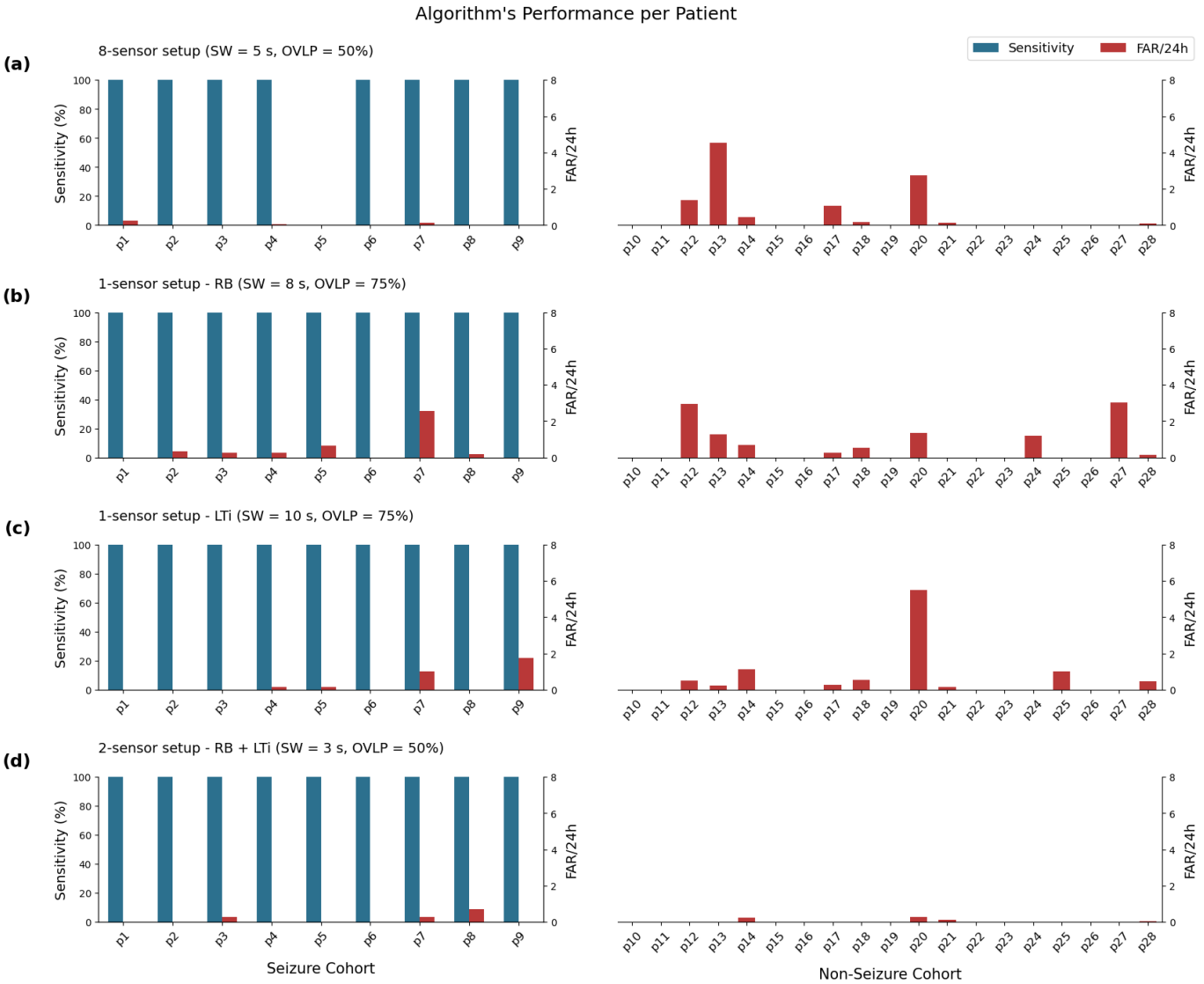

Fig. 3. Algorithm's performance across both cohorts when using the best-performing models from each sensor setup: **(a)** 8-sensor setup with data from all muscles; **(b)** 1-sensor setup with data from the right biceps brachii; **(c)** 1-sensor setup with data from the left tibialis anterior; **(d)** 2-sensor setup with data from both the right biceps brachii and the left tibialis anterior. Abbreviations: false alarm rate (FAR), sliding window (SW), overlap (OVLP), right biceps brachii (RB), left tibialis anterior (LTi).

who met the inclusion criteria for each cohort. No signal pre-screening or patient selection processes were performed, despite periods of sensor detachment or noisy conditions from loose electrodes. For evaluation on the non-seizure cohort, the machine learning algorithm was fixed and frozen, run offline with predefined hyperparameters, and none of the patients in this cohort were involved in model development or selection.

No sensor-related adverse events were observed during the study. Skin irritation from prolonged electrode wear occurred in only one patient who stayed in the EMU for over three weeks, leading to an early termination of data acquisition. The irritation was comparable to that caused by the ECG electrode used in clinical practice. In a real-world application, daily hygiene routines, such as showering and moisturizing, would likely mitigate such effects by reducing sweat and salt accumulation on the skin. Beyond this case, three patients decided to end their participation due to discomfort from sensor wear. The data recorded from these patients prior to ending their participation were still included in the analyses. In one case (p20), the discomfort was attributed to electrode gel liquefying as a result of skin temperature, which caused electrodes to constantly shift from their placement and adhere to clothing. These factors likely contributed to the degraded signal quality and may explain the elevated FAR observed in this patient compared to the rest of the cohort.

On eight occasions, data recording was unexpectedly interrupted mid-session, resulting in data loss. In these occasions, Cometa's EMG and Motion Tools software displayed an error message related to USB connection speed, despite following the manufacturer's recommendations for cable and port use. These incidents were thus classified as device deficiencies, accounting for 1.28% of the times a recording session was initiated across the 50 recruited patients. In a real-world application, however, the algorithm would ideally run on a

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mobile application, with sensors transmitting data wirelessly to a smartphone for real-time analysis, reducing the likelihood of problems associated with wired connections.

Overall, our algorithm showed promising performance in detecting GTCS/FBTCS in an EMU. With eight sensors, the best-performing model achieved 90.9% sensitivity with 0.05 micro FAR/24h over 1,359.6 h of continuous recordings from the seizure cohort, and 0.56 over 1,744.18 h of continuous recordings from the non-seizure cohort. Using a single sensor increased sensitivity to 100% but also raised micro FAR/24h to 0.51 (RB) and 0.28 (LTi) in the seizure cohort, and 0.50 (RB) and 0.62 (LTi) in the non-seizure cohort. This suggests that a single-sensor setup is more susceptible to misclassifying localized bursts of activity as seizures, likely due to the absence of contextual information from other muscles. Our results also show inter-patient variability in FAR, indicating poor algorithm performance in specific cases, particularly on patients who accounted for a disproportionately large share of false alarms. The most common cause of high FAR in these patients was poor signal quality due to loose electrodes. Other factors included abrupt upper-limb activities, such as prolonged pillow shaking and repetitive jolt-like movements. A possible solution to reduce FAR in these cases involves the firing power postprocessing applied in our method. Although the threshold L was automatically optimized in this study, it could be individually adjusted in later applications to favor sensitivity or specificity according to personal preferences and needs. However, despite this variability, the median FAR/24h across patients ranged from 0 (LTi) to 0.28 (RB) in the seizure cohort, and remained 0 in the non-seizure cohort, compared to 0 in both cohorts when using eight sensors. These values suggest that for most patients, FAR remained low even with a minimal sensor setup, supporting the feasibility of reducing the number of sensors in our approach. A single-sensor configuration could substantially improve usability and practicality by increasing patient comfort and simplifying setup.

Importantly, the single- and dual- sensor setups in this study were primarily evaluated on data from the RB and LTi muscles, as these were the only sites where sensors remained attached throughout all seizures. Although results for the other muscles are reported in the supplementary material (Tables S5–S8), these were obtained by training the algorithm on different patient subsets from the seizure cohort, based on the number of seizures successfully recorded from each muscle (i.e., when sensors did not detach). Consequently, the algorithm was tested on fewer seizures and over shorter recording durations, reflecting only an approximate estimate of performance. Further validation is therefore required to determine the optimal sensor placement. However, the results do suggest that comparable performance could still be achieved with alternative muscles. This would enable personalized sensor placement to accommodate patient preferences and needs. For example, selecting a location that is easily concealed under clothing, generates fewer false alarms, or maximizes comfort. Patients with higher FAR may benefit from multi-sensor configurations, as our results suggest that additional sensors could reduce false alarms. Intermediate setups with two or four sensors, rather than eight, could potentially provide a balance between comfort and precision. We demonstrated this by evaluating our algorithm on data from the RB and LTi muscles combined. This dual-sensor setup achieved the highest performance among all the explored setups in the study, reaching 100% sensitivity with a micro FAR/24h of 0.12 and 0.06 on the seizure and non-seizure cohorts, respectively.

Additionally, to identify the top biologically meaningful contributing features, we examined their gain (i.e., average loss improvement) as an indicator of their contribution to the algorithm's prediction. Overall, the STD of jerk along the ACC z-axis was the most influential feature across all sensor setups. One possible explanation is that it captures variability in acceleration changes, particularly in the vertical direction, which may be more pronounced during the clonic phase of GTCS/FBTCS due to abrupt shifts in movement patterns. In the 8-sensor setup, the features with the highest gain values were primarily extracted from the anterior deltoids and upper trapezii, likely explaining the missed seizure when both trapezius sensors detached. This suggests that the classifier relied more heavily on specific muscles and that the lack of characteristic seizure patterns on the signals it considered most informative might have led to the FN; hence the lower sensitivity with the 8-sensor setup. In contrast, the sensors on the RB and LTi remained stably attached during all seizures; though there were detachment periods during interictal states, they consistently captured physiologically meaningful signals during seizures, as opposed to background noise, thereby providing the classifier with informative input. Other important features in the 8-sensor setup included the STD of jerk, ACC- and sEMG-based amplitude measures (MAV, RMS), ACC median frequency, sEMG peak frequency, and the STD of the sEMG wavelet coefficients in the 62–250 Hz band, which might reflect the high-frequency muscle activation expected during the tonic phase due to sustained muscle stiffening. In the single-sensor setups, the most influential features were ACC-derived measures such as interquartile range, STD of jerk, RMS, and median frequency, as well as sEMG-derived features including entropy across the 2–8 Hz, 8–62 Hz, and 62–250 Hz bands, and the high-to-low frequency ratio. The latter aligns well with previous findings [20] and is thought to characterize the transition from the tonic to the clonic phase. Finally, the features with the highest gain in the dual-sensor setup included the STD of jerk, maximum jerk, and sEMG band power across the 2–8 Hz, 8–62 Hz, and 62–250 Hz bands.

The results presented in this work are comparable to previous Phase 2-3 studies targeting GTCS/FBTCS using unimodal or multimodal devices (Table VI). All sensor setups achieved $>$90% sensitivity, with the majority maintaining slightly lower micro FAR/24h than has been reported in earlier studies using sEMG-based [12], [13] or FDA-cleared [13], [15] devices. Interestingly, both single-sensor setups showed favorable results compared to sEMG-based monitors placed on the biceps brachii [12], [13] in terms of both sensitivity and micro FAR/24h, but they were outperformed in the latter metric by wrist-worn devices [14], [24], [31]. While our detection latency was longer than the $<$10 s reported in sEMG-based studies [12], [13], it was comparable to that observed when ACC is incorporated [14]–[16], [24]. This is likely due to the

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TABLE VI
COMPARISON WITH EXISTING LITERATURE

Study	Device [modality]	Evaluation method	Patients, n	Seizures, n	Testing time, h	Sensitivity, %	Micro FAR/24h	Median latency, s
Beniczky, et al. (2013) [14]	Epi-Care wristband [ACC]*	Real-time	20	39	4,878	89.7	0.2	33 (BTC)‡
Halford, et al. (2017) [13]	SPEAC biceps monitor [sEMG]*	Real-time	199	46	9,237	76	2.52	7.38 (BTC)
Beniczky, et al. (2018) [12]	EDDI biceps monitor [sEMG]*	Real-time	20	32	3,735.5	93.8	0.67	9 (BTC)
Onorati, et al. (2021) [15]	Embrace* and E4 wristbands [ACC, EDA]	Independent test set	18	31	6,357	94	0.57	37.46 (clinical)
Gharbi, et al. (2024) [16]	Biometric shirt [ECG, ACC]	NCV	42	66	8,067	84.8	0.69†	25.5 (BTC)
Vakilina, et al. (2024) [31]	Samsung watch [ACC, GYR, PPG]	LOPO CV	9	15	681	87	0.21	Not reported
		Independent test set	19	0	1,116	-	0.28	-
		Independent test set	8	3	680	100	0.25	Not reported
Spahr, et al. (2025) [24]	E4 wristband [ACC]	CV	37	54	4,922	98	0.18	Not reported
		Independent test set	347	49	31,479	96	0.13	26 (BTC)
This study	Eight miniature sensors [sEMG, ACC]	LOPO NCV	9	11	1,359.6	90.9	0.05	22.5 (BTC)
		Independent test set	19	0	1,744.18	-	0.56	-
	One miniature biceps sensor [sEMG, ACC]	LOPO NCV	9	11	1,359.6	100	0.51	32 (BTC)
		Independent test set	19	0	1,744.18	-	0.50	-
	One miniature leg sensor [sEMG, ACC]	LOPO NCV	9	11	1,359.6	100	0.28	26.5 (BTC)
		Independent test set	19	0	1,744.18	-	0.62	-
	Two miniature sensors [sEMG, ACC]	LOPO NCV	9	11	1,359.6	100	0.12	22 (BTC)
		Independent test set	19	0	1,744.18	-	0.06	-

Abbreviations: surface electromyography (sEMG), accelerometry (ACC), electrocardiography (ECG), electrodermal activity (EDA), gyroscope (GYR), photoplethysmography (PPG), cross-validation (CV), nested cross-validation (NCV), leave-one-patient-out (LOPO), bilateral tonic-clonic (BTC).

*Seizure detection devices that either are, or have been, commercially available.

†Micro FAR/24h was calculated based on the total number of false alarms (FP = 235) and total recording time reported in the study.

‡Mean latency is reported instead of the median.

applied postprocessing method, which introduces an inherent detection delay as a result of the moving average calculation. Nevertheless, the measured latencies remain within the time-frame considered adequate for timely seizure intervention (<3 min) in the context of SUDEP prevention [11].

The limitations of this study are centered around three main points. First, our dataset was relatively small compared to others used in existing literature. This limitation primarily arose from using a single data acquisition system, which allowed recordings from only one patient at a time, unlike in multicenter studies [12]–[15], [24], [31]. Naturally, the small dataset size posed a classic challenge for machine learning in high-dimensional feature spaces, which we addressed through implicit feature selection (e.g., L_1 regularization), strategies to control model complexity (e.g., L_2 regularization, increased *gamma*), and rigorous evaluation procedures (e.g., NCV). Though the results show promise in FBTCS/GTCS detection, we must recognize that our small, homogeneous patient cohort might limit our algorithm's generalizability to broader patient populations, and that larger studies (e.g., Phases 3 and 4) must be conducted to fully assess its clinical utility. To mitigate this limitation, we evaluated our algorithm on patients who did not have any clinical epileptic seizures recorded by the sensors, providing an additional estimate of the model's FAR. Testing on independent datasets, even when cross-validation methods are employed, is becoming imperative in algorithm validation to provide unbiased evidence of model performance, with the most recent studies increasingly adopting this method [24], [31]. Second, the limited 10-hour battery life of the sensors presented a significant challenge. Following the loss of two seizures due to battery depletion, an intensive sensor replacement schedule of three times per day was implemented. Although effective for the study, this approach is not feasible in real-world applications, emphasizing the need for longer battery duration. Rapid advances in technology could potentially address this limitation, with similar sensors now offering

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18 hours of battery life [32]. Third, the study was conducted in **(1)** a hospital setting, which inherently restricts the physical activity of patients, and **(2)** in a single EMU, which might not reflect the variability seen in different centers. Consequently, FARs observed in this controlled environment may underestimate those likely to occur in outpatient settings, where unregulated artifacts are more frequent and could trigger more false alarms. Therefore, future research will focus on large-scale, real-time evaluation of the algorithm across multiple centers and in real-life conditions.

V. Conclusion

Wearable devices that monitor biosignals offer a promising approach for out-of-clinic seizure detection, with the potential to improve patient safety and independence in real-world settings. In this study, we developed a seizure detection algorithm for GTCS and FBTCS based on sEMG and ACC signals recorded with miniaturized wearable sensors. Our approach demonstrated promising performance with eight-, two-, and single-sensor setups in an EMU, laying the foundation for the use of miniaturized technology in the context of seizure detection. Future research will focus on validating the developed algorithm in outpatient environments.

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